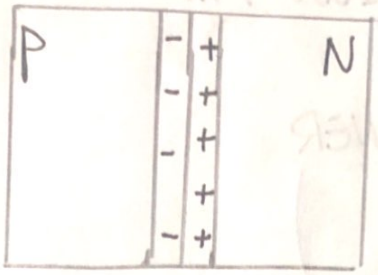
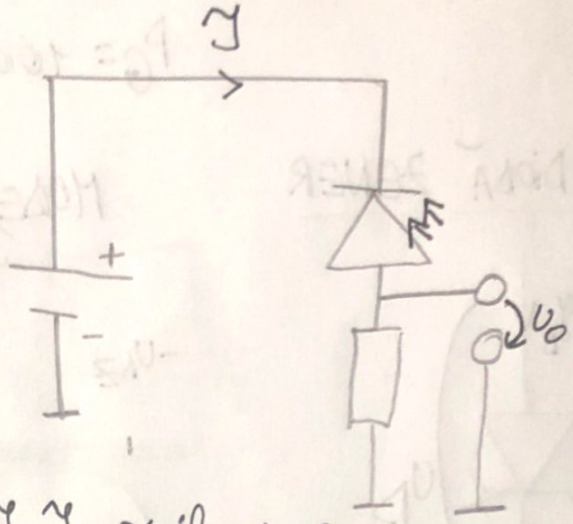
Curs LEEA

FOTODIODĂ



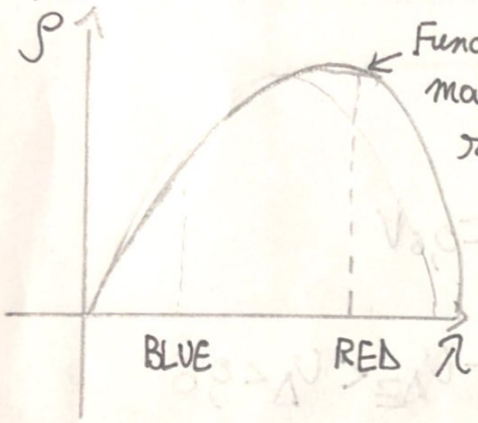




$I = I_S \approx$ iluminare

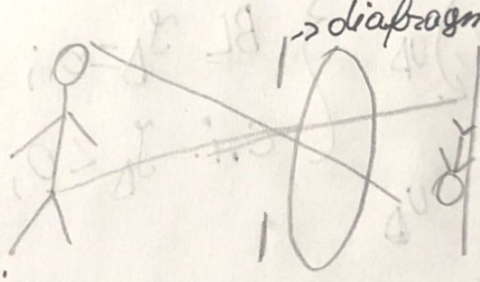
$U_0 = I \cdot R \approx$ iluminare

Eficienta fotodiodei



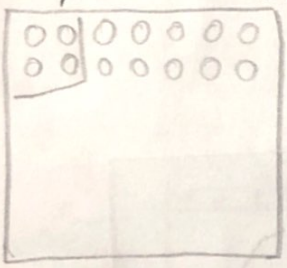
Exemple: senzor telefon

Camera foto - atâtia pixeli = atâtea diode



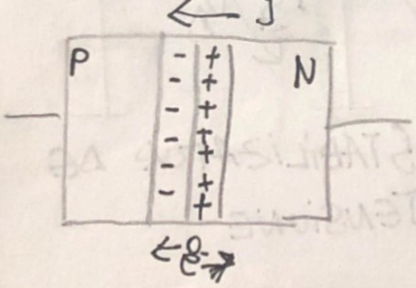
→ o matrice de fotodiode care se încarcă cu energie în funcție de luminozitatea care bate înspre ele

GB → Prin filtre iese
RG → culoarea în acea zonă de pixeli

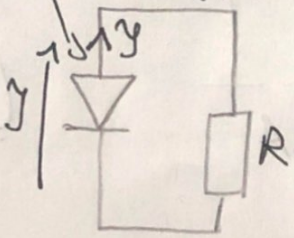


Diafragma se ocupă să nu supraturezze imaginea

SOLAR CELL



tensiune
sursă de ~~curent~~ de max. 0,6V



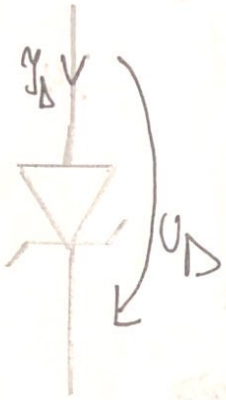
- Le legăm în serie ca să obținem valori mai mari de ~~curent~~ tensiune

Scanned with CamScanner

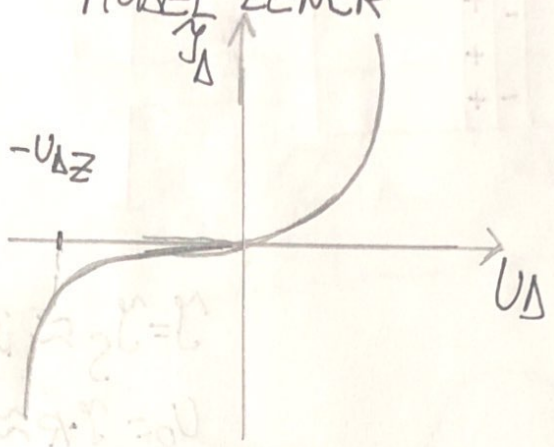
η 10% 20% $P_S = 1 kW/m^2$ (putere solară) 1KWh

$P_G = 100W - 200W/m^2$

DIODĂ ZENER

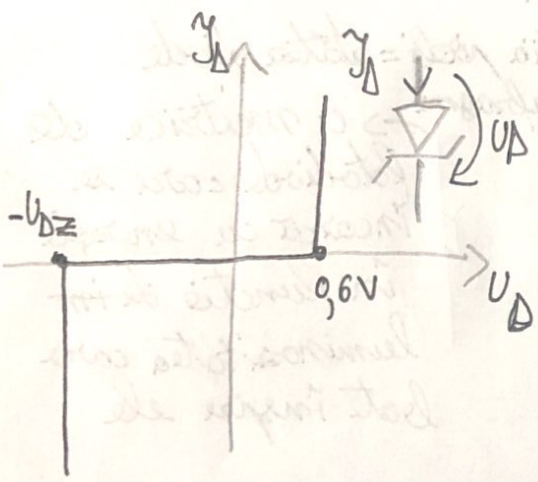


MODEL ZENER



U_{DZ} - precizat (tensiunea de străpungere)
VOLȚI - 10VOLȚI

APROXIMARE ZENER



$$\begin{cases} \text{C.D} & \gamma_\Delta > 0;\ U_\Delta = 0{,}6V \\ \text{BL} & \gamma_\Delta = 0;\ -U_{\Delta Z} < U_\Delta < 0{,}6 \\ \text{C.I.} & \gamma_\Delta < 0;\ U_\Delta = -U_{DZ} \end{cases}$$

Exemplu: $\gamma = ?$

$U_{\Delta Z} = 5V$

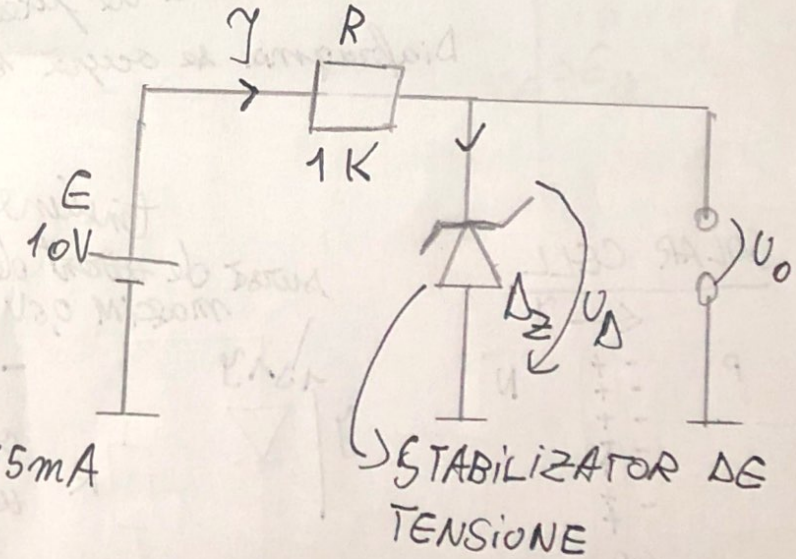
PP Δ - CONDUCȚIE INVERSĂ

$U_\Delta = 5V$ $\gamma > 0$

$E = R \cdot \gamma + U_\Delta \Rightarrow \gamma = \frac{10V - 5V}{1K} = 5mA$

$U_0 = U_\Delta = 5V$

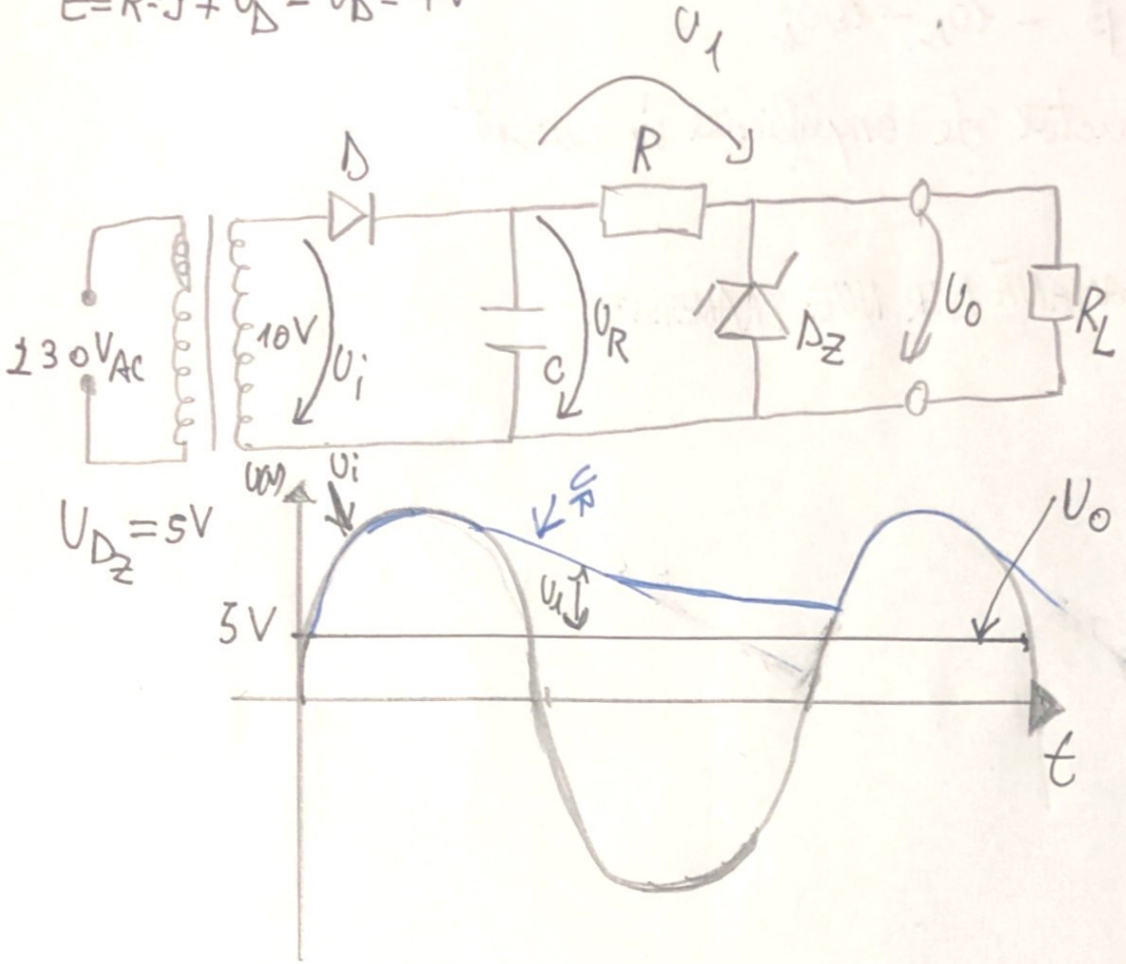
Dacă $E = 12 \Rightarrow U_0 = 5V$ (tot 5V)



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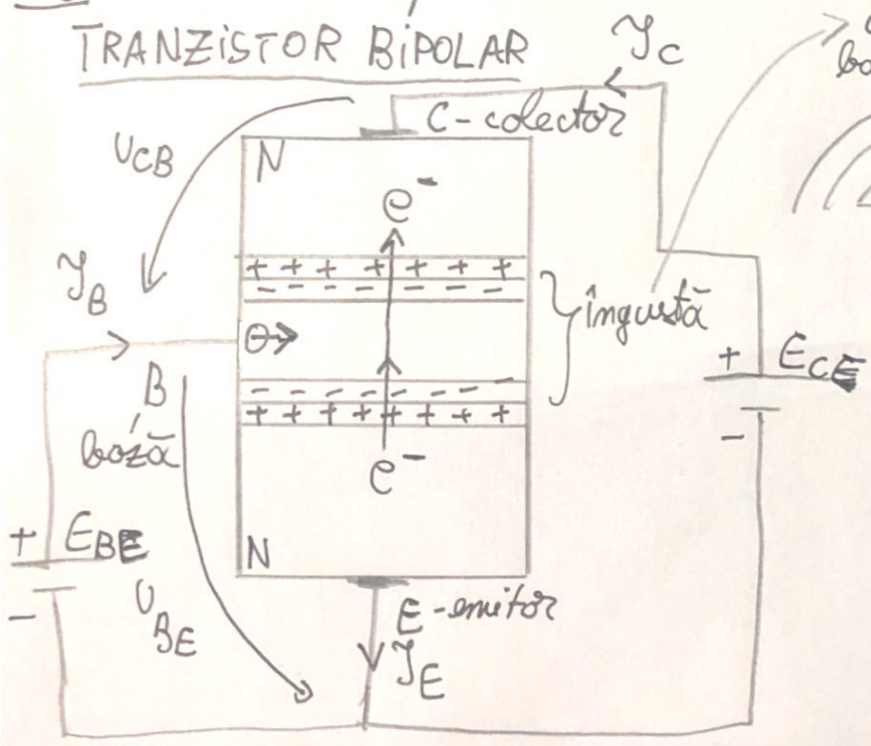
$U_0 = 4V = U_D \Rightarrow$ D-BL $\Rightarrow \gamma = 0$

$E = R \cdot \gamma + U_D = U_D = 4V$

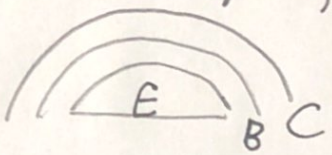


Obs: Diodele nu pot amplifica un semnal

TRANZISTOR BIPOLAR



ca electronii să sară peste bariera de potențial



Dacă $E_{CE} > E_{BE} \Rightarrow U_{BE} > 0$

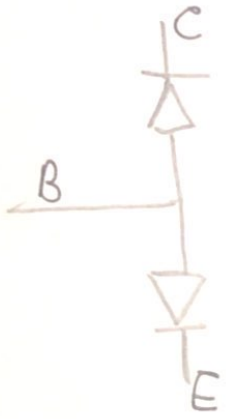
BE - CONDUCȚIE DIRECTĂ

Dacă $U_{CB} = E_{CE} - E_{BE}$ CB - INVERS BLOCATĂ

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$$\begin{cases} I_E = I_B + I_C \\ I_C = \beta I_B ; \ \beta - 10_{\text{ș}} - 100_{\text{ș}} \end{cases}$$

↳ factor de amplificare în curent



SEAMĂNĂ DAR NUE TRANZISTOR

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